

1.

To find the probability that one ball is red, one is white, and one is black we use combinations

* Red balls : 6

White balls : 4

Black balls : 8

Total balls = $6 + 4 + 8$
 $= 18$

Number of balls drawn : 3

The total number of ways to choose 3 balls from 18 is given by ${}^{18}C_3$

$${}^{18}C_3 = \frac{18 \times 17 \times 16}{3 \times 2 \times 1} = 816$$

We need exactly 1 ball of each color

Ways to pick 1 red ball : ${}_6C_1 = 6$

Ways to pick 1 white ball ${}_4C_1 = 4$

Ways to pick 1 black ball ${}_8C_1 = 8$

Total favorable outcomes = $6 \times 4 \times 8 = 192$

Final probability

$$P(1 \text{ Red}, 1 \text{ White}, 1 \text{ Black}) = \frac{\text{Favorable Outcomes}}{\text{Total Outcomes}}$$

$$P = \frac{192}{816}$$

Simplifying the fraction (dividing both by 48)

$$P = \frac{4}{17}$$

Answer : The probability is $\frac{4}{17}$ (0.235 approximately)

2.

Let $P(A)$, $P(B)$, and $P(C)$ be the probabilities that students A, B, and C solve the problem respectively

$$\bullet P(A) = \frac{1}{2}$$

$$\bullet P(B) = \frac{1}{3}$$

$$\bullet P(C) = \frac{1}{4}$$

The probability that each student fails to solve the problem is

$$P(A') = 1 - \frac{1}{2} = \frac{1}{2}$$

$$P(B') = 1 - \frac{1}{3} = \frac{2}{3}$$

$$P(C') = 1 - \frac{1}{4} = \frac{3}{4}$$

Since the students work independently we multiply their individual failure probabilities

$$P(\text{None solve}) = P(A') \times P(B') \times P(C')$$

$$P(\text{None solve}) = \frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} = \frac{6}{24} = \frac{1}{4}$$

The problem is solved if ~~as~~ at least one person gets it right:

$$P(\text{solved}) = 1 - P(\text{None solve})$$

$$P(\text{solved}) = 1 - \frac{1}{4} = \frac{3}{4}$$

Answer: The probability that the problem will be solved is $\frac{3}{4}$ (or 0.75)

3. Analyze the days in a leap year.

A leap year has 366 days

$366 \div 7 = 52$ weeks and 2 extra days

Determine the possible combination for the 2 extra days

The two consecutive extra days can be

Monday and Tuesday

Tuesday and Wednesday

Wednesday and Thursday

Thursday and Friday

Friday and Saturday

Saturday and Sunday

Sunday and Monday

There are 7 total possibilities for the extra days. Out of these, 2 possibilities

(~~Saturday / Sunday~~ and Monday) (Pairs 6 and 7 contain a Sunday).

$$\text{Probability} = \frac{\text{Favorable Outcomes}}{\text{Total Outcomes}}$$

$$= \frac{2}{7}$$

4. We first need to convert the inclusive class intervals into continuous classes (exclusive form) and find the class marks (x_i)

We adjust the classes by subtracting 0.5 from the lower limit and adding 0.5 to the upper limit.

Class (x)	Continuous class	Frequency (f_i)	Class Mark (x_i)	$f_i \cdot x_i$	Cumulative Freq. (cf)
0-4	-0.5-4.5	4	2	8	4
5-9	4.5-9.5	1	7	7	5
10-14	9.5-14.5	1	12	12	6
15-19	14.5-19.5	10	17	170	16
20-24	19.5-24.5	3	22	66	19
25-29	24.5-29.5	25	27	135	24
30-34	29.5-34.5	3	32	96	27

35-39	34.5-39.5	0	37	0	27
40-44	39.5-44.5	1	42	42	28

Total

$$N = 28$$

$$\sum f_i X_i \\ = 536$$

The mean ($\bar{x}$) is calculated using the formula

$$\bar{x} = \frac{\sum f_i X_i}{N} = \frac{536}{28} \approx 19.14$$

Total Frequency (N) : 28

$$\text{Median Class Position: } \frac{N}{2} = \frac{28}{2} = 14$$

The cumulative frequency just greater than 14 is 16, which corresponds to the class 14.5 - 19.5

Median Formula

$$\text{Median} = L + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h$$

where,

- L (Lower limit of median class) = 14.5
- cf (Cumulative frequency of preceding class) = 6

- f (Frequency of median class) = 10
- h (class width) = 5

$$\text{Median} = 14.5 + \left(\frac{14 - 6}{10} \right) \times 5$$

$$\begin{aligned} \text{Median} &= 14.5 + \left(\frac{8}{10} \right) \times 5 = 14.5 + 4 \\ &= 18.5 \end{aligned}$$

Final results

$$\text{Mean} : \approx 19.14$$

$$\text{Median} : 18.5$$

5. We will use the ~~step~~-Deviation Method
 Let the assumed mean (A) be 42.5 and
 the class width (h) be 5
 We define $u_i = \frac{x_i - A}{h}$

Class Interval	Midpoint (x_i)	Frequency (f_i)	$u_i = \frac{x_i - 42.5}{5}$	$f_i u_i$	$f_i u_i^2$
20-25	22.5	3	-4	-12	48
25-30	27.5	15	-3	-45	135
30-35	32.5	45	-2	-90	180
35-40	37.5	57	-1	-57	57
40-45	42.5	50	0	0	0
45-50	47.5	36	1	36	36
50-55	52.5	25	2	50	100
55-60	57.5	9	3	27	81
Total		$N = 240$	$\sum f_i u_i$ $= -91$	$\sum f_i u_i$ $= -91$	$\sum f_i u_i^2$ $= 637$

The formula for standard deviation (σ) using step-deviation is

$$\sigma = h \times \sqrt{\frac{\sum f_i u_i^2}{N} - \left(\frac{\sum f_i u_i}{N}\right)^2}$$

Substituting the values

$$\sigma = 5 \times \sqrt{\frac{637}{240} - \left(\frac{-91}{240}\right)^2}$$

$$\sigma = 5 \times \sqrt{2.6542 - (-0.3792)^2}$$

$$\sigma = 5 \times \sqrt{2.6542 - 0.1438}$$

$$\sigma = 5 \times \sqrt{2.5104}$$

$$\sigma = 5 \times 1.5844$$

$$\sigma \approx 7.922$$

Final Answer:

The standard deviation of the given distribution is approximately 7.92.